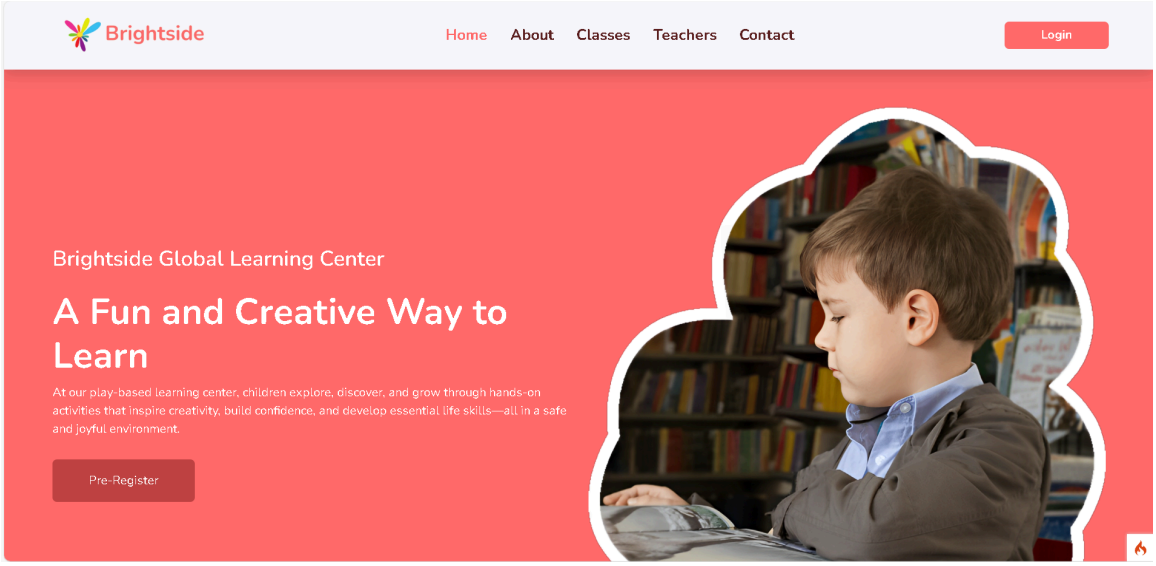
EDGE: UX/UI Case Study Report — Phase 2

1. System Overview

System Title: EDGE: Educational Development and Guardian Engagement – A Multi-Feature Platform for Play Schools

The EDGE system aims to strengthen communication and engagement among schools, vc 0oikjhteachers, parents, and students. It automates attendance tracking, tuition management, announcements, and learning features while maintaining an engaging platform for both teachers and learners. The system promotes accessibility, transparency, and efficiency in managing school operations.

Target Users: Administrators, Teachers, Students, Parents/Guardians, and Staff

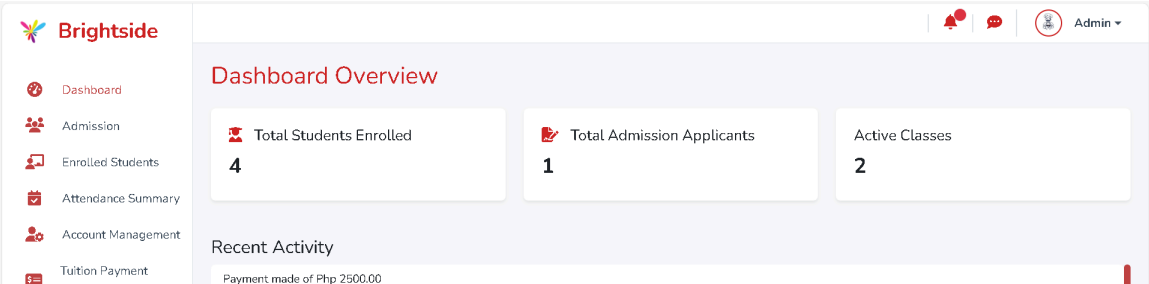


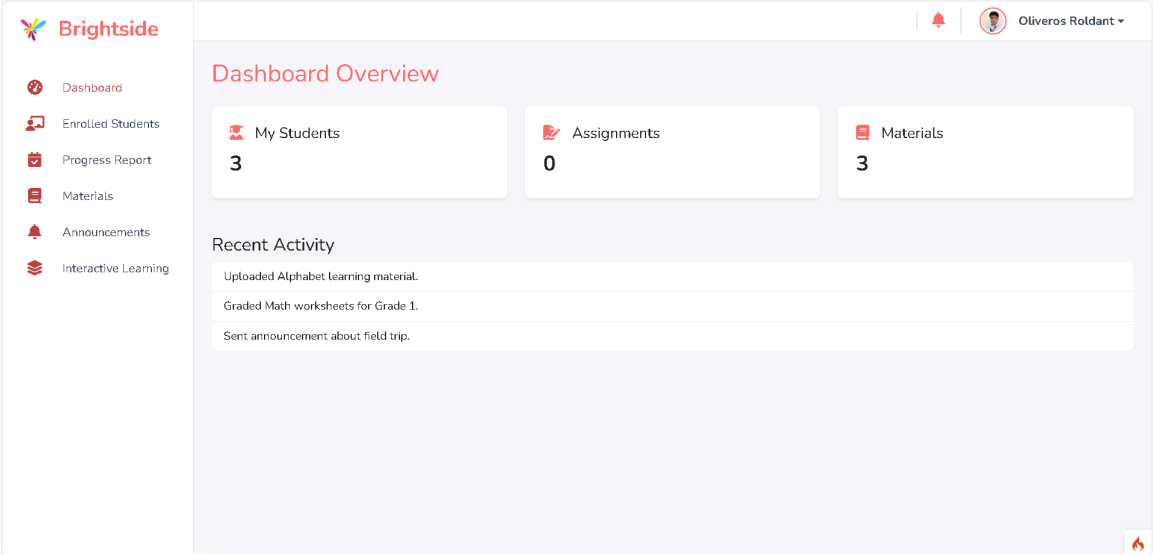
2. Design Evaluation (Nielsen's 10 Usability Heuristics)

The system was assessed using Nielsen's 10 usability heuristics to identify key strengths and weaknesses in its user interface and overall usability.

Heuristic	Strengths	Weaknesses
1. Visibility of System Status	Dashboards show real-time data such as enrolled students, active classes, and payments.	Some modules (attendance, forms) do not display confirmation messages after submission or scanning.

2. Match Between System and Real World	Uses familiar words like 'Students,' 'Attendance,' and 'Payments.'	Homepage, About, Classes, and Teachers show identical content, which confuses users.
3. User Control and Freedom	Admin customization settings allow easy adjustment of system colors and themes.	Some actions lack cancel or undo options, especially in forms and attendance scanning.
4. Consistency and Standards	Menu structure and icon styles remain consistent across all user roles.	Inconsistent color schemes across pages — some use red and white, while others add blue.
5. Error Prevention	Structured input forms minimize user mistakes.	Lack of real-time validation or warning messages may cause unnoticed input errors.
6. Recognition Rather Than Recall	Sidebar icons make navigation easier.	Some buttons and inputs lack labels or descriptive tooltips.
7. Flexibility and Efficiency of Use	Dashboards display summaries for quick decision-making.	No filters or search shortcuts for large datasets.
8. Aesthetic and Minimalist Design	Overall design is clean and not overloaded with information.	Overuse of bright red creates visual fatigue, and white-on-white text fields reduce readability.
9. Help Users Recognize, Diagnose, and Recover from Errors	Notification icons and announcements indicate system events.	No specific error indicators or guides to help users recover from mistakes.
10. Help and Documentation	The interface is simple enough for first-time users.	The system lacks built-in help or documentation to guide new users.

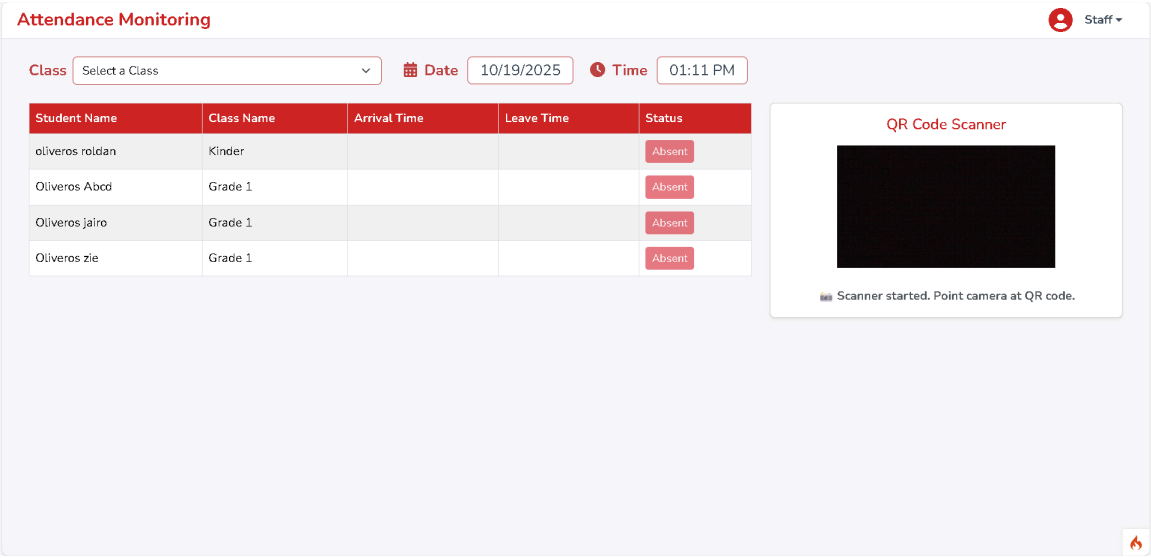


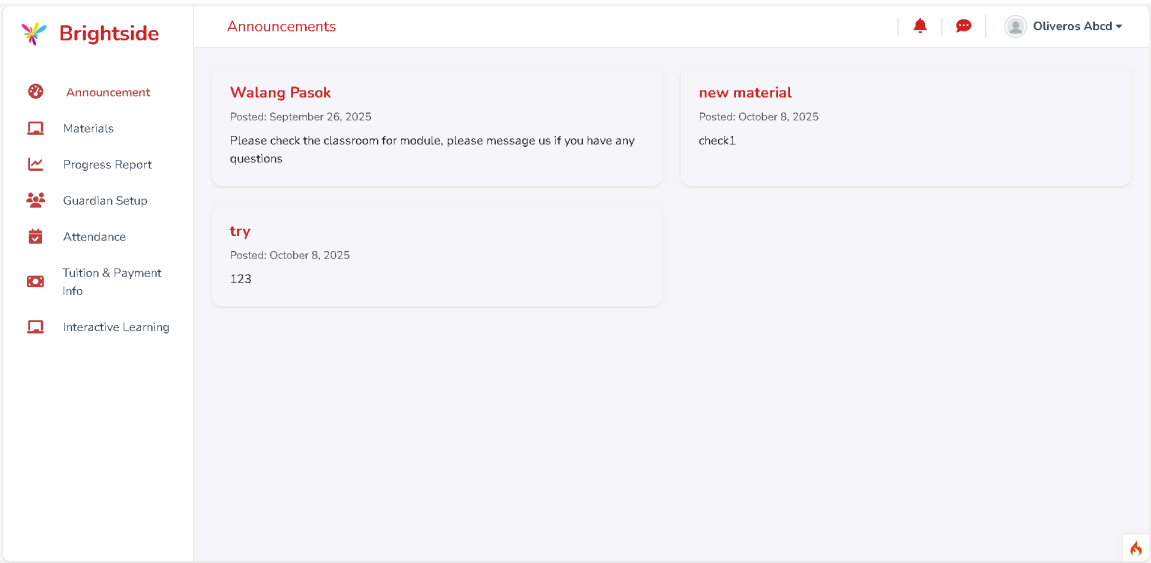


3. User Experience Observations

Navigation is generally easy because of the consistent sidebar layout and recognizable icons. However, users notice that some pages repeat identical content, leading to confusion. The combination of bright red and white backgrounds causes eye strain and reduces focus. The contact and form pages have light input boxes that blend into the white background, reducing readability. Teachers and admin dashboards look almost identical, which may cause confusion about user roles.

On a positive note, the inclusion of interactive learning games for teachers and students increases engagement and promotes a fun learning experience. The attendance monitoring feature works, but it looks plain and lacks clear feedback after scanning. Color contrast, spacing, and role-specific differentiation are areas for improvement.





4. Recommendations

- Use softer tones or complementary colors to reduce visual strain and improve readability.
- Provide unique content for each webpage to make navigation meaningful and informative.
- Add confirmation messages or success indicators when users perform important actions.
- Differentiate dashboard themes by role (e.g., color accents for Admin, Teacher, Student).
- Maintain consistent color schemes and spacing for a cohesive look across all modules.
- Include error validation messages for incorrect form entries or failed actions.
- Improve layout hierarchy using shadows, dividers, and spacing for clarity.

• Add a Help or Tutorial section to support new users during onboarding.

[Insert Screenshot: Suggested Redesign Mockups Here]

Redesign — Phase 3